

The Anchor Protocol:

A Formally Verified Control Plane for Local Agent Swarms

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Abstract

As AI agents transition from isolated copilots to collaborative, autonomous swarms, local development environments face a crisis of coordination and trust. Existing tools designed for human-driven, static infrastructures lack the primitives required to prevent dynamic agents from corrupting shared state or squatting on ephemeral ports.

This paper introduces the **Anchor Protocol**, a cryptographic and semantic identity framework built into the Port Daddy daemon. We detail the protocol’s evolution from symmetric MACs to asymmetric Ed25519 signatures and multi-hop delegation chains (inspired by Macaroons [4]). Furthermore, we present the formal verification of the protocol’s design using the **ProVerif** symbolic analyzer [1] and the mathematical proof of its implementation, demonstrating immunity to algorithm confusion [10], impersonation, and timing side-channels.

Keywords: distributed systems security, formal verification, capability-based security, multi-agent coordination, symbolic protocol analysis, ProVerif, JWT security, Ed25519

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1 Introduction: The “Local Swarm” Problem

In a multi-agent development environment, agents operate concurrently to read files, spin up test servers, and execute commands. This introduces three critical threat vectors on `localhost`:

1. **Port Squatting (The “Ghost in the Harbor”)**: If Agent A crashes, a malicious or malfunctioning Agent B can bind to Agent A’s previously assigned port, intercepting traffic meant for A.
2. **Resource Contention**: Agents fighting over shared resources (e.g., a SQLite database file) without a centralized locking mechanism cause state corruption.
3. **Privilege Escalation**: A “Reviewer” agent delegated by a “Coder” agent should not possess the rights to initiate production deployments, yet local environments rarely enforce principle-of-least-privilege between processes.

To solve this, Port Daddy introduces the concept of a **Harbor**—a zero-trust, namespace-isolated execution environment where agents must prove their identity and capabilities. Our approach builds upon foundational work in capability-based security by Dennis and Van Horn [7] and the principle of least privilege articulated by Saltzer and Schroeder [15].

2 The Anchor Protocol Architecture

The Anchor Protocol defines how agents authenticate to a Harbor and to each other. It issues **Harbor Cards** (highly constrained JSON Web Tokens).

2.1 Phase 1: Symmetric Pinning

Early versions of the protocol relied on HS256 (HMAC-SHA256). The primary vulnerability in JWTs is “Algorithm Confusion” (CVE-2015-9235 [11], CVE-2023-48238 [13]), where an attacker modifies the token header to `{"alg": "none"}` or symmetric HS256 while using an asymmetric public key as the HMAC secret.

Property 2.1 (Algorithmic Pinning). The Port Daddy Verifier employs strict **Algorithmic Pinning**. It entirely ignores the user-provided `alg` header and explicitly forces the underlying crypto library to evaluate the signature using the pre-determined algorithm. This approach is recommended by the JWT Best Current Practices draft [14].

2.2 Phase 2: Asymmetric Identity (Ed25519)

To support decentralized Agent-to-Agent (A2A) communication without sharing HMAC secrets, the protocol transitioned to **Ed25519 (EdDSA)** [8, 9]. Ed25519 provides fast, deterministic signatures with small keys and signatures, making it ideal for local agent communication.

Definition 2.1 (Harbor Key Hierarchy). The Port Daddy Daemon acts as the Root Certificate Authority (CA). Each Harbor is assigned a unique Ed25519 keypair. The Daemon issues Harbor Cards signed by the Harbor’s private key. Agents use the Harbor’s public key (broadcast via Port Daddy’s ambient discovery service) to verify tokens presented by peers.

2.3 Phase 3: Stigmergic Delegation (The “Biscuit” Pattern)

Port Daddy v4 introduces multi-hop delegation. When Agent A spawns Agent B, it does not need to request a new token from the Daemon. Instead, it uses **Offline Attenuation**, inspired by Macaroons [4].

Definition 2.2 (Offline Attenuation). Agent A takes its existing Harbor Card, appends a new set of restricted capabilities (e.g., ["db:read"] reduced from ["db:read", "db:write"]), and signs the *entire chain* with its own ephemeral private key.

The Harbor verifies the Daemon’s root signature, Agent A’s delegation signature, and mathematically ensures that Agent B’s capabilities are a strict subset of Agent A’s.

3 Related Work

3.1 Formal Verification of Security Protocols

Formal verification of cryptographic protocols has matured significantly over the past two decades. Following the seminal work by Lowe [5] on authentication hierarchies and the Dolev-Yao model [6], several automated tools have been developed:

- **ProVerif** [1]: An automatic symbolic protocol verifier supporting an unbounded number of sessions, used to verify TLS 1.3 [17], Signal [16], and WireGuard.
- **Tamarin** [18]: A state-of-the-art protocol verifier that supports equational properties and inductive proofs, used for analyzing 5G authentication [19].
- **CryptoVerif** [2]: A computationally-sound protocol verifier that provides proofs in the computational model rather than the symbolic model.

Our work builds on these foundations, applying established verification techniques to the novel domain of local multi-agent coordination.

3.2 Capability-Based Authorization

Capability-based security dates back to Dennis and Van Horn’s seminal 1966 paper on programming semantics for multiprogrammed computations [7]. Recent work by Birgisson et al. [4] introduced Macaroons, which use chained HMACs for decentralized delegation with contextual caveats. The Anchor Protocol’s delegation chains (Section 5.3) are inspired by Macaroons but adapted for JWT-based identity in local development environments.

3.3 JWT Security

JSON Web Tokens have been the subject of extensive security analysis. Algorithm confusion attacks were first systematically studied by McLean [10], with subsequent vulnerabilities documented in CVE-2015-9235 [11] (HS256/RS256 confusion), CVE-2018-1000531 [12] (“none” algorithm), and CVE-2023-48238 [13] (generic algorithm confusion). Our protocol implements the defense recommended by the IETF JWT Best Current Practices [14]: strict algorithm whitelisting and key separation.

4 Formal Verification Strategy

Following the industry standard set by AWS (s2n-tls [20]) and Microsoft (Project Everest [22]), we decoupled the verification of the *protocol design* from the verification of the *implementation*.

4.1 Symbolic Analysis with ProVerif

To prove the protocol’s logical soundness against a Dolev-Yao adversary [6] (an attacker who controls the entire network), we modeled the Anchor Protocol in **ProVerif** [1]. ProVerif represents protocols as Horn clauses and uses resolution to prove security properties for an unbounded number of sessions.

Theorem 4.1 (Injective Agreement). For the multi-hop delegation chain, ProVerif confirmed that:

$$\text{Accepted}(b, h, \text{cap}) \implies (\text{IssuedRoot}(a, h, \text{cap}) \wedge \text{Delegated}(a, b, \text{cap})) \vee \text{IssuedRoot}(b, h, \text{cap}) \quad (1)$$

This *injective agreement* property [5] guarantees that capabilities cannot be escalated and trust is perfectly transitive.

4.2 Implementation Verification

A secure protocol is useless if the implementation contains buffer overflows or timing leaks. We extracted the critical JWT parsing and cryptographic comparison logic into a verified core.

Property 4.1 (Memory Safety). Using the **Kani** Rust Verifier [21], we performed bounded model checking to prove that our parsing logic never panics or accesses out-of-bounds memory, regardless of the input payload.

Property 4.2 (Side-Channel Resistance). To prevent timing attacks where an adversary guesses signatures byte-by-byte [23], we implemented a custom constant-time byte comparator. Kani formally verified that this comparator is branch-free regarding the contents of the secret arrays.

5 Verification Algorithms

In this section, we present the verification algorithms in pseudocode format, following the notation used in protocol verification literature [1, 5]. These algorithms correspond to the formal ProVerif models in Appendix A.

5.1 Phase 1: Symmetric HS256 with Algorithm Pinning

Algorithm 1 presents the secure verification procedure that is immune to algorithm confusion attacks (CVE-2015-9235 [11]).

```

1 Require: Pinned Algorithm: hs256
2 Require: Master Key: k_master (secret)
3
4 function Daemon.IssueToken(agent_id, harbor_id, capability):
5     jti <- GenerateNonce()
6     msg <- Encode(agent_id, harbor_id, jti, capability)
7     sigma <- HMAC-SHA256(msg, k_master)
8     emit Issued(agent_id, harbor_id, jti, capability)
9     return (hs256, msg, sigma)
10
11 function Harbor.VerifyToken(tok, expected_harbor):
12     (alg_header, msg, sigma) <- tok
13     // CRITICAL: Ignore alg_header, pin to HS256
14     if HMAC-SHA256(msg, k_master) != sigma:
15         return _|_ // Invalid signature
16     (a, h, j, cap) <- Decode(msg)
17     if h != expected_harbor:
18         return _|_ // Wrong harbor
19     emit Accepted(a, h, j, cap)
20     return (a, h, j, cap)

```

Listing 1: Algorithm 1: Secure Harbor Verification (HS256 with Algorithm Pinning)

Security Property: The algorithm ignores the user-provided alg_{header} (line 15), preventing algorithm confusion attacks where an attacker sets $alg = \text{“none”}$ or attempts HS256/RS256 confusion [10].

5.2 Phase 2: Asymmetric Ed25519

Algorithm 2 presents the Ed25519-based verification. This phase removes the need for shared HMAC secrets between distributed Harbors.

```

1 Require: Harbor Keypair: (sk_harbor, pk_harbor)
2 Require: Key Distribution Channel: c_key (private)
3
4 function Daemon.IssueToken(agent_id, harbor_id, capability):
5     sk_h <- Receive(c_key) // Get harbor's secret key
6     jti <- GenerateNonce()
7     msg <- Encode(agent_id, harbor_id, jti, capability)
8     sigma <- Ed25519.Sign(msg, sk_h)
9     emit Issued(agent_id, harbor_id, jti, capability)
10    return (msg, sigma)
11
12 function Harbor.VerifyToken(tok, pk_h, expected_harbor):
13     (msg, sigma) <- tok
14     if not Ed25519.Verify(msg, sigma, pk_h):
15         return |_ // Invalid signature
16     (a, h, j, cap) <- Decode(msg)
17     if h != expected_harbor:
18         return |_ // Wrong harbor
19     emit Accepted(a, h, j, cap)
20    return (a, h, j, cap)

```

Listing 2: Algorithm 2: Asymmetric Harbor Verification (Ed25519)

Security Property: The secrecy of sk_{harbor} and the unforgeability of Ed25519 [8] ensure that only the legitimate Daemon can issue valid Harbor Cards.

5.3 Phase 3: Multi-hop Delegation

Algorithm 3 presents the delegation chain verification, inspired by Macaroons [4].

```

1 Require: Daemon Public Key: pk_daemon
2 Require: Subset Relation: subseteq on capabilities
3
4 function Agent.Delegate(token_parent, sk_parent, agent_child, cap_sub):
5     (msg_parent, sigma_parent) <- token_parent
6     (_, agent_parent, harbor, cap_parent) <- Decode(msg_parent)
7     if cap_sub not subseteq cap_parent:
8         return |_ // Cannot escalate capabilities
9     msg_delegate <- Encode_delegation(msg_parent, sigma_parent,
10     agent_child, cap_sub)
11     sigma_delegate <- Ed25519.Sign(msg_delegate, sk_parent)
12     emit Delegated(agent_parent, agent_child, cap_sub)
13     return (msg_delegate, sigma_delegate)
14
15 function Harbor.VerifyChain(chain, pk_root):
16     tok_root <- chain[0]
17     (msg_0, sigma_0) <- tok_root
18     if not Ed25519.Verify(msg_0, sigma_0, pk_root):
19         return |_ // Invalid root signature

```

```

19   cap_root <- ExtractCaps(msg_0)
20   pk_current <- pk_root
21   for i <- 1 to |chain| - 1:
22       (msg_i, sigma_i) <- chain[i]
23       if not Ed25519.Verify(msg_i, sigma_i, pk_current):
24           return |_ // Invalid delegation signature
25       cap_i <- ExtractCaps(msg_i)
26       if cap_i not_subseteq cap_root:
27           return |_ // Capability escalation detected
28       pk_current <- ExtractPubkey(msg_i)
29   emit Accepted(agent_final, harbor, cap_final)
30   return T

```

Listing 3: Algorithm 3: Delegation Chain Verification

Security Property: The subset check on line 7 and line 22 ensures *capability attenuation*—delegated capabilities can only be restricted, never expanded. Combined with Theorem 4.1, this guarantees transitive trust.

6 Implementation: Rust Core

The following Rust implementation demonstrates the verified cryptographic core:

```

1  /// Harbor Token Verification Core
2  ///
3  /// This module implements constant-time cryptographic verification
4  /// to prevent timing side-channel attacks.
5
6  use subtle::ConstantTimeEq;
7  use ed25519_dalek::{VerifyingKey, Signature, Verifier};
8
9  /// A Harbor Card containing identity and capabilities
10 #[derive(Debug, Clone)]
11 pub struct HarborCard {
12     pub agent_id: String,
13     pub harbor_name: String,
14     pub capabilities: Vec<String>,
15     pub issued_at: u64,
16     pub expires_at: u64,
17 }
18
19 /// Constant-time byte comparison to prevent timing attacks
20 ///
21 /// # Security
22 /// This function is verified to be branch-free with respect to
23 /// the contents of the input arrays.
24 pub fn constant_time_verify(a: &[u8], b: &[u8]) -> bool {
25     if a.len() != b.len() {
26         return false;
27     }
28     a.ct_eq(b).into()
29 }
30
31 /// Verifies an Ed25519-signed Harbor Card
32 pub fn verify_harbor_card(
33     token: &[u8],
34     signature: &Signature,
35     harbor_pubkey: &VerifyingKey,

```

```

36 ) -> Result<HarborCard, VerifyError> {
37     // Verify signature first
38     harbor_pubkey
39         .verify(token, signature)
40         .map_err(|_| VerifyError::InvalidSignature)?;
41
42     let card = parse_token(token)?;
43
44     // Verify not expired
45     let now = current_timestamp();
46     if card.expires_at < now {
47         return Err(VerifyError::Expired);
48     }
49
50     Ok(card)
51 }
52
53 /// Verifies a delegation chain
54 pub fn verify_delegation_chain(
55     chain: &DelegationChain,
56     root_pubkey: &VerifyingKey,
57 ) -> Result<HarborCard, VerifyError> {
58     // Verify root signature
59     let root = &chain.hops[0];
60     root_pubkey
61         .verify(&root.token, &root.signature)
62         .map_err(|_| VerifyError::InvalidRootSignature)?;
63
64     let root_cap = parse_capabilities(&root.token)?;
65
66     // Verify each delegation hop
67     for window in chain.hops.windows(2) {
68         let parent = &window[0];
69         let child = &window[1];
70
71         let parent_pk = extract_pubkey(parent)?;
72
73         parent_pk
74             .verify(&child.token, &child.signature)
75             .map_err(|_| VerifyError::InvalidDelegationSignature)?;
76
77         let child_cap = parse_capabilities(&child.token)?;
78         if !is_subset(&child_cap, &root_cap) {
79             return Err(VerifyError::CapabilityEscalation);
80         }
81     }
82
83     parse_token(&chain.hops.last().unwrap().token)
84 }
85
86 fn is_subset(a: &[String], b: &[String]) -> bool {
87     a.iter().all(|cap| b.contains(cap))
88 }

```

Listing 4: Rust Implementation — Constant-Time Verification

7 Security Analysis

7.1 Threat Model

We assume a **Dolev-Yao adversary** [6] who:

- Controls the entire network
- Can intercept, modify, and inject messages
- Cannot break cryptographic primitives (cryptography is perfect)
- May corrupt legitimate agents (but not all simultaneously)

This is the standard threat model for symbolic protocol analysis [1].

7.2 Security Properties

Property	Verified	Tool	Reference
Master Key Secrecy	Yes	ProVerif	[1]
Algorithm Confusion Immunity	Yes	ProVerif	[10]
Authentication (Injective Agreement)	Yes	ProVerif	[5]
Delegation Integrity	Yes	ProVerif	[4]
Memory Safety	Yes	Kani	[21]
Constant-Time Comparison	Yes	Kani	[23]

Table 1: Verified Security Properties

7.3 Limitations

1. **PID Binding:** The “Ghost in the Harbor” scenario requires binding tokens to the requesting PID, which is handled at the OS level.
2. **Local Transport Security:** On multi-user systems, local unix sockets or loopback TLS should be used to prevent local bearer token sniffing.

8 Conclusion

The Anchor Protocol elevates local multi-agent development from a state of “hope-based security” to “math-based security.” By combining:

- Symbolic protocol proofs (ProVerif [1])
- Memory-safe implementation (Rust/Kani [21])
- Capability-based delegation (inspired by Macaroons [4])

Port Daddy provides a formally verified control plane capable of safely orchestrating the next generation of autonomous AI swarms.

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A Full ProVerif Models

For completeness, we include the full ProVerif models used for verification. These correspond to the algorithms in Section 5.

A.1 Phase 1: Symmetric HS256

```

1 (* Port Daddy: Harbor Card v1 (HS256) Formal Model *)
2 type id. type key. type jti. type capability. type alg_type.
3 const hs256_alg: alg_type. const none_alg: alg_type.
4 free c: channel.
5
6 fun hs256(bitstring, key): bitstring.
7   reduc forall m: bitstring, k: key;
8     check_hs256(m, k, hs256(m, k)) = true.
9
10  reduc forall m: bitstring, k: key;
11    verify_generic(m, hs256_alg, hs256(m, k), k) = true;
12    forall m: bitstring, k: key;
13      verify_generic(m, none_alg, m, k) = true.
14
15 fun encode_token(id, id, jti, capability): bitstring [data].
16
17 event Issued(id, id, jti, capability).
18 event Accepted(id, id, jti, capability).
19
20 free master_key: key [private].

```

```

21 query attacker(master_key).
22
23 query a: id, h: id, j: jti, cap: capability;
24   event(Accepted(a, h, j, cap)) ==> event(Issued(a, h, j, cap)).
25
26 let Daemon(k: key, a_id: id, h_id: id, j: jti, cap: capability) =
27   let msg = encode_token(a_id, h_id, j, cap) in
28   let signature = hs256(msg, k) in
29   event Issued(a_id, h_id, j, cap);
30   out(c, (hs256_alg, msg, signature)).
31
32 let SecureHarbor(k: key, expected_h: id) =
33   in(c, (alg_header: alg_type, msg: bitstring, signature: bitstring))
34   ;
35   if check_hs256(msg, k, signature) = true then
36     let encode_token(a: id, h: id, j: jti, cap: capability) = msg in
37     if h = expected_h then
38       event Accepted(a, h, j, cap).
39
40 process
41   new agent_id: id; new harbor_id: id;
42   new token_jti: jti; new secret_cap: capability;
43   (!Daemon(master_key, agent_id, harbor_id, token_jti, secret_cap) |
    !SecureHarbor(master_key, harbor_id))

```

Listing 5: harbor_card_v1_refined.pv